

## OPEN-SOURCE &amp; FELLOWSHIPS

## Featured Research & Interactive Demo

Open-source initiatives and flagship research architectures indexing knowledge and powering autonomous decision-making.

Google Summer of Code Fellow (2024 &amp; 2025)

Open Source (R Foundation)

### AI Scholar — Open-Source Hybrid RAG Platform [↗](#)

Solo-architected an open-source literature search and synthesis platform indexing **15M+ scientific research papers**. Combines dense semantic vector retrieval (FAISS), lexical BM25 search, and Neo4j graph traversal with meta-learning to dynamically personalize retrieval strategies.

- ▶ **+40% Retrieval Relevance:** Boosted NDCG@10 relevance by 40% over baseline on a 200-query held-out benchmark with citation-derived relevance judgments.
- ▶ **Attributed Chain-of-Thought:** Built reasoning chains with explicit source attribution and calibrated uncertainty quantification (Monte Carlo dropout, evidential deep learning).
- ▶ **Dynamic Knowledge Graph:** Implemented named entity recognition (spaCy, BERT-NER) and relation extraction to automatically link concepts, researchers, and scientific methodologies.

#### [▶ Interactive Hybrid RAG Architecture Sandbox](#)

Benchmark Query: Quantum Error Correction &amp; Surface Codes

##### 1. DENSE FAISS VECTOR

**0.942 Cosine Similarity**

Top-k (k=10) dense passages from 15M+ corpus via custom physics embeddings

##### 2. SPARSE BM25 LEXICAL

**16.85 BM25 Score**

Exact lexical tokens: 'surface code', 'threshold', 'gauge theory', 'stabilizer'

##### 3. NEO4J KNOWLEDGE GRAPH

**3-hop Graph Path**

Entity (Lattice Gauge Theory) → Subgraph (Google Quantum AI) → Method (VQE / Surface

Code)

Meta-Learning Fusion: **+42.4% NDCG@10 vs baseline**      **Evidential Variance  $\sigma^2 = 0.041$  (High Confidence)**

Surface codes exhibit fault-tolerant thresholds (~1% per physical gate) under toric topology. In lattice gauge simulations, mapping gauge invariants onto syndrome measurements allows variational quantum eigensolvers (VQE) to mitigate error propagation across 50+ qubits.

[Fowler et al., Phys. Rev. A]

[Mejo et al., Google Quantum AI Initiative]

[Aaronson et al., ACM STOC]

Quantitative Finance & Execution

Braverock LLC

## Microsecond Foundation Models & Deep RL Trading System

Production reinforcement-learning trading framework and transformer forecasting pipeline processing 1B+ quote-data points per day at microsecond latency.

- ▶ **Live Trading Alpha:** Delivered a 6.2 Sharpe ratio (Sortino 8.4, max drawdown <1.9%) over a continuous 24-month live deployment window, executing billions of dollars across sub-second to multi-month holding periods.
- ▶ **Distributed Foundation Models:** Directed distributed pretraining and instruction fine-tuning pipelines for domain-specific 7B–14B foundation models using DeepSpeed ZeRO-3 and custom CUDA memory-mapped tokenizers (-38% per-step wall-clock latency).
- ▶ **RAG on 10M+ Filings:** Vectorized and indexed SEC filings, earnings calls, and broker research reports for real-time signal extraction.

Fintech Transformer Engine

Cushion AI

## CushionNLP — Transformer-Powered Automated Negotiation System

Proprietary BERT-based transformer system designed for automated financial negotiation and transaction categorization across 300+ financial institutions.

- ▶ **\$2M+ Customer Savings:** Drove a 30% increase in successful bank-fee negotiations and \$2M+ in customer savings in year one.
- ▶ **80M+ Transactions Categorized:** Graph neural networks and ensemble classifiers achieved 95% accuracy while cutting processing latency by 50% compared to legacy rules engines.

## ACADEMIC PEDIGREE

## Education & Credentials

Theoretical foundation rooted in quantum information, theoretical computer science, and complexity theory.

### Postdoctoral Fellowship, Theoretical Computer Science

— Massachusetts Institute of Technology (MIT)

2010 — 2012

Faculty Advisor: **Dr. Scott Aaronson**. Research concentrated on quantum complexity theory, computational limits of quantum systems, and foundational computer science.

### Ph.D. in Physics — Harvard University

2007 — 2010

Specialization: **Quantum information and complexity theory**. Completed degree requirements at age 21.

### B.S. in Physics and Computer Science — Harvard University

2004 — 2007

### Top Secret / Sensitive Compartmented Information (TS/SCI) — U.S. Government

Active Clearance

# Resume / Curriculum Vitae

Verbatim reproduction of Christopher Mejo's executive resume. Formatted for screen reading and optimized for clean 2-page print or PDF export.

## CHRISTOPHER MEJO Ph.D.

### AI Research Scientist — LLM & Reinforcement Learning Systems

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## SUMMARY

AI research scientist and engineer with 15+ years taking machine learning systems from research prototype to production, holding a Ph.D. in Physics from Harvard (quantum information and complexity theory, completed at 21) and postdoctoral training at MIT under Scott Aaronson. Core focus: LLM systems — retrieval-augmented generation, fine-tuning, agentic reasoning — and reinforcement learning for sequential decision-making. Solo-architected AI Scholar, an open-source RAG platform indexing 15M+ research papers (Google Summer of Code Fellow, 2024 and 2025). Shipped production LLM and RL systems in finance and a federated computer-vision system in healthcare. Comfortable owning systems end-to-end in small teams.

## PROFESSIONAL EXPERIENCE

### Primary Employment

#### Senior Data Scientist — Braverock LLC

(Nov 2021–Present)

- Architected a reinforcement-learning trading system (deep Q-networks, actor-critic methods) for multi-asset portfolio optimization, improving Sharpe ratio 23% over prior strategy on microsecond-latency data, validated across both backtested and live trading.
- Built transformer-based market-prediction models for financial time series, deployed in a pipeline processing 1B+ quote-data points/day at microsecond-scale latency.
- Built a RAG system over 10M+ financial documents (SEC filings, earnings calls, research reports) for signal extraction, using custom embeddings and a production vector database.
- Directed distributed pretraining and instruction fine-tuning pipelines for domain-specific 7B–14B parameter foundation models (DeepSpeed ZeRO-3, custom CUDA memory-mapped tokenizers), reducing per-step wall-clock latency 38%.
- Implemented dynamic Heterogeneous Graph Attention Networks (HAN) modeling multi-asset cross-correlations across tick, intraday, and multi-week horizons, improving regime-shift detection accuracy 19%.
- Designed and deployed an enterprise crypto basis-yield strategy delivering an out-of-sample, beta-neutral Sharpe ratio of 2.63 on \$60M+ in institutional AUM.
- Developed proprietary quantitative strategies achieving a 6.2 Sharpe ratio (Sortino 8.4, max drawdown <1.9%) over a continuous 24-month live deployment window, executing billions of dollars in trading volume across holding periods from sub-second to multi-month.
- Strategy scope spanned quantitative multi-exchange, multi-asset market making; treasury management; regime-aware risk-managed momentum; and multi- and single-asset statistical arbitrage / structural mean-reversion market-neutral strategies.

#### Senior Data Scientist — Cushion AI

(Apr 2021–Oct 2021)

- Built CushionNLP, a proprietary BERT-based transformer system for bank-fee negotiation, driving a 30% increase in successful negotiations and \$2M+ in additional annual customer savings.
- Architected a transaction-categorization system (ensemble methods, graph neural networks) achieving 95% accuracy across 80M+ transactions from ~300 financial institutions, cutting processing time 50% versus the prior rule-based (non-ML) system.

#### Research Data Scientist — RapidRads AI

(2019–Feb 2021)

- Built a computer-vision system (EfficientNet-B7 backbone) for COVID-19 detection from chest X-rays, achieving 0.972 ROC-AUC with 96.5% sensitivity and 98.2% specificity (1.8% false-positive rate, 3.5% false-negative rate) under differential privacy ( $\epsilon = 1.2$ ,  $\delta = 10^{-5}$ ) across a federated-learning deployment spanning 4 medical institutions.

- Combined multiple CNN architectures via ensemble methods with Grad-CAM visualization and uncertainty quantification for clinical validation.

**Senior Data Scientist & Cybersecurity Researcher — Confiant***(2017–2019)*

- Built threat-detection models (Isolation Forest, One-Class SVM, LSTM autoencoders) and GraphSAGE-based botnet-detection models identifying coordinated attacks across 1B+ daily ad requests.

**CEO & Lead Architect — Cronode***(2011–2017)*

- Built a recommendation engine (TF-IDF, cosine similarity, collaborative filtering) and custom Solr ranking configuration.

Concurrent Research & Fellowships

Visiting Research Scientist (part-time) — Sandia National Laboratories — Quantum Computing & High Energy Physics, Joint Google Quantum AI Initiative

(2020–2024)

- Senior researcher in a joint research initiative with Google Quantum AI developing variational quantum eigensolvers for lattice gauge theory calculations.
- Designed quantum error correction protocols (surface codes) and resource-estimation frameworks for near-term quantum devices (Qiskit, Cirq).
- Ran distributed quantum many-body simulations (50+ qubits) on SLURM-managed clusters; contributed to DARPA and IARPA research proposals.
- Active Top Secret/Sensitive Compartmented Information (TS/SCI) security clearance.

Google Summer of Code Fellow, 2024 & 2025 — AI Scholar Project — R Foundation for Statistical Computing

(2024–2025)

- Architected AI Scholar ([github.com/cmejo/AI\\_Scholar](https://github.com/cmejo/AI_Scholar)), combining dense (FAISS), sparse (BM25), and graph-based (Neo4j) retrieval for scientific literature search across 15M+ research papers.
- Built an adaptive retrieval system using meta-learning to personalize search strategy, improving relevance 40% (NDCG@10) over baseline on a 200-query held-out set with citation-derived relevance judgments.
- Implemented chain-of-thought reasoning over the RAG pipeline with explicit source attribution and calibrated uncertainty (Monte Carlo dropout, evidential deep learning).
- Built a dynamic knowledge-graph construction pipeline using named entity recognition (spaCy, BERT-NER) and relation extraction to connect concepts, authors, and methodologies.

EDUCATION

Postdoctoral Fellowship, Theoretical Computer Science — Massachusetts Institute of Technology

(2010–2012)

Faculty Advisor: Dr. Scott Aaronson.

Ph.D. in Physics — Harvard University

(2007–2010)

Quantum information and complexity theory; completed at 21.

B.S. in Physics and Computer Science — Harvard University

(2004–2007)

TECHNICAL SKILLS

**Languages:** Python, C, C++, R, Assembly, Java, JavaScript

**ML / DL:** PyTorch, TensorFlow, JAX, Transformers, Reinforcement Learning (DQN, actor-critic, MAML)

**LLM / RAG:** Retrieval-augmented generation (dense, sparse, and graph-based retrieval), FAISS, BM25, Neo4j knowledge graphs, custom embedding models, vector databases, LangChain, LangGraph, vLLM, named entity recognition (spaCy, BERT-NER), chain-of-thought and agentic reasoning, uncertainty quantification (Monte Carlo dropout, evidential deep learning), retrieval evaluation (NDCG, MRR)

**MLOps:** MLflow, Kubeflow, Apache Airflow, Docker, Kubernetes, Ray, Weights & Biases

**Quantum Computing:** Qiskit, Cirq

**Distributed / HPC:** SLURM, multi-GPU distributed training